

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – Multimodal Mini-Demo (Text-to-Image + Speech-to-Text/Text-to-Speech)

Course Code:	CSA65	Course Title:	Generative AI and Large Scale Models
CO Assessed:	CO4 – Precision (BL3, BL4,BL5)	Assessment:	Assessment Tool 3 - Multimodal Mini-Demo (Text-to-Image + Speech-to-Text/Text-to-Speech)
Weightage:	10% of CIA	Total Marks:	2 * 50= 100
CO4 Statement:	<i>Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.(BL3, BL4,BL5)</i>		
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Section / Batch:	CSE-AI	Date:	27-08-2026

Instructions to Students

- Answer any 2 questions. Each question carries 50 marks. Total: 100 marks.
- Develop and execute the assigned **Python program** using Text-to-Image, Speech-to-Text, or Text-to-Speech.
- Explain the **working principle and program workflow**.
- Demonstrate the program with **suitable inputs and outputs**.
- Use appropriate **libraries/APIs** and include basic **error handling**.
- Show the **execution output/screenshots**.
- For integrated tasks, demonstrate the complete **multimodal workflow**.
- Include a **workflow/architecture diagram or comparison table**.
- Mention a **real-world application** of the implemented solution.
- Explain the **strengths and limitations** of the approach.
- Include proper **references/citations** for the libraries, APIs, models, or research sources used.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

SECTION A : Multimodal Mini-Demo (Text-to-Image + Speech-to-Text/Text-to-Speech)

1. Modify a text-to-image program to allow the user to repeatedly generate images until they select exit.

1. Title

AI Text-to-Image Generator with Repeat and Exit Option

2. Objective

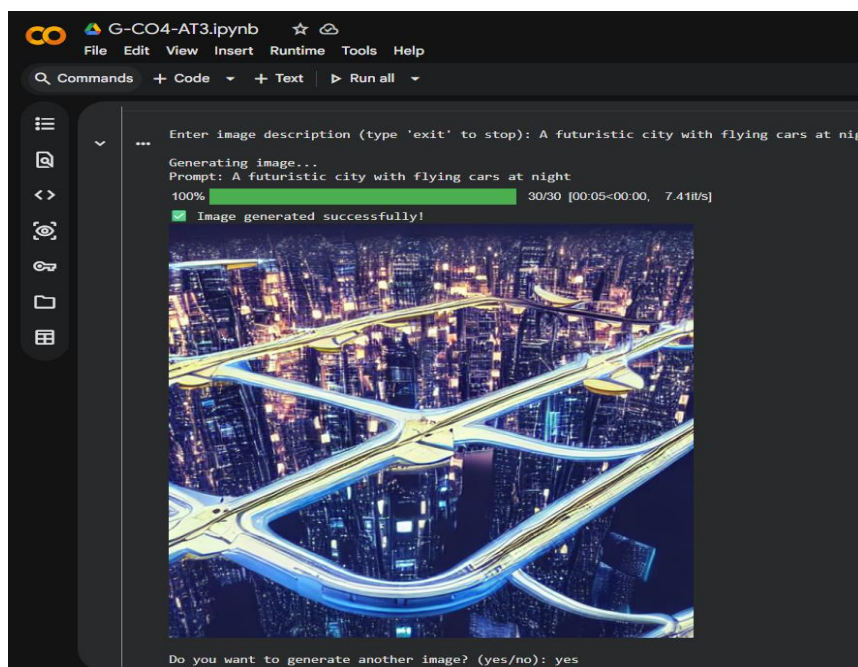
The objective of this program is to develop a Text-to-Image application that accepts a text description from the user and generates a corresponding AI-created image. The application allows the user to generate multiple images repeatedly until the Exit option is selected.

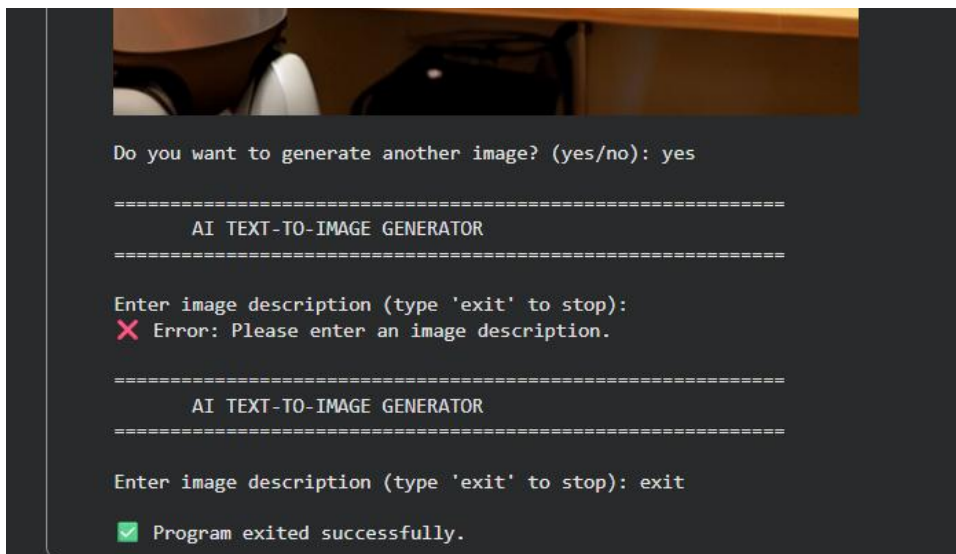
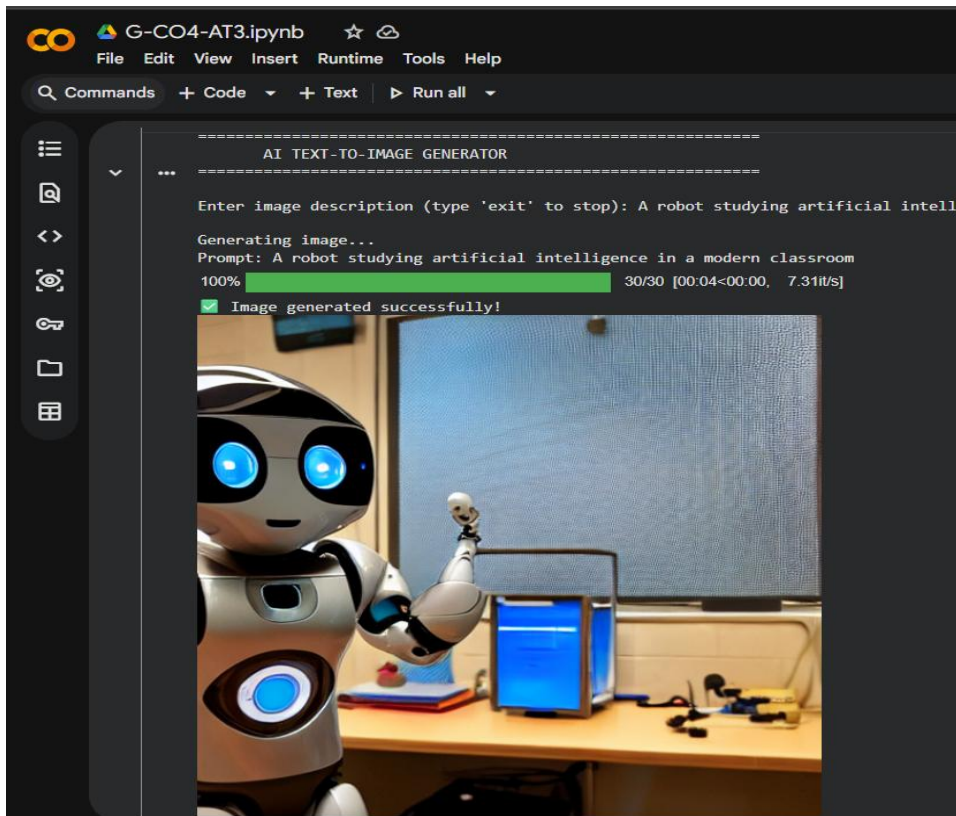
3. Technologies Used

- Python
- Google Colab
- PyTorch
- Hugging Face Diffusers
- Stable Diffusion

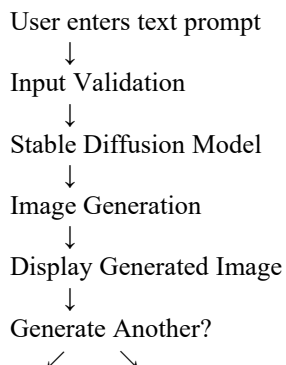
4. Working Principle

The user enters a description of the desired image. The text prompt is passed to the Stable Diffusion Text-to-Image model, which processes the prompt and generates an image based on its meaning. After displaying the image, the program asks whether the user wants to generate another image. If the user selects **Yes**, a new prompt can be entered. If the user selects **No** or **Exit**, the program terminates.





5. Workflow



Yes No
↓ ↓
New Prompt Exit

6. Implementation

The program was implemented using Python and the Stable Diffusion model. The model converts the textual description into an image. Input validation was included to handle empty prompts, and an exit condition was provided to stop the application.

The program uses a loop so that the user can generate multiple images without restarting the application.

7. Sample Inputs

- A futuristic city with flying cars at night
- A robot studying in a modern classroom
- A beautiful waterfall surrounded by green mountains

8. Error Handling

The application handles invalid inputs such as:

- Empty image description
- Invalid continuation choice
- Exit command

For an empty prompt, the program displays an error message and asks the user to enter a valid description.

9. Advantages

- Generates images directly from natural language.
- Allows repeated image generation.
- Simple and interactive workflow.
- Can be used for creative and visual applications.

10. Result and Conclusion

The Text-to-Image application successfully converts user-provided text descriptions into AI-generated images. The repeat-and-exit functionality allows multiple images to be generated during a single execution. The system also provides basic input validation and error handling.

2. Write a complete mini-demo in which the user speaks a description, the program converts it into text, and the recognized text is used as the prompt for image generation.

1. Title

Voice-to-Image AI Generator Using Speech-to-Text and Text-to-Image

2. Objective

The objective of this project is to develop an integrated multimodal application in which the user speaks an image description. The application converts the speech into text and uses the recognized text as the prompt for an image-generation model.

3. Technologies Used

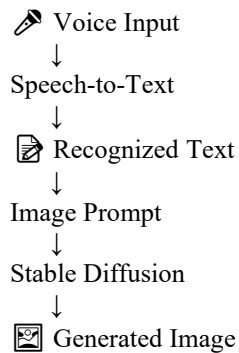
- Python
- Google Colab
- SpeechRecognition
- Gradio
- Stable Diffusion
- PyTorch

4. Working Principle

The user provides an image description through the microphone. The Speech-to-Text component recognizes the spoken words and converts them into text.

The recognized text is then directly passed to the Stable Diffusion Text-to-Image model. The model generates an image based on the spoken description, which is displayed through the Gradio interface.

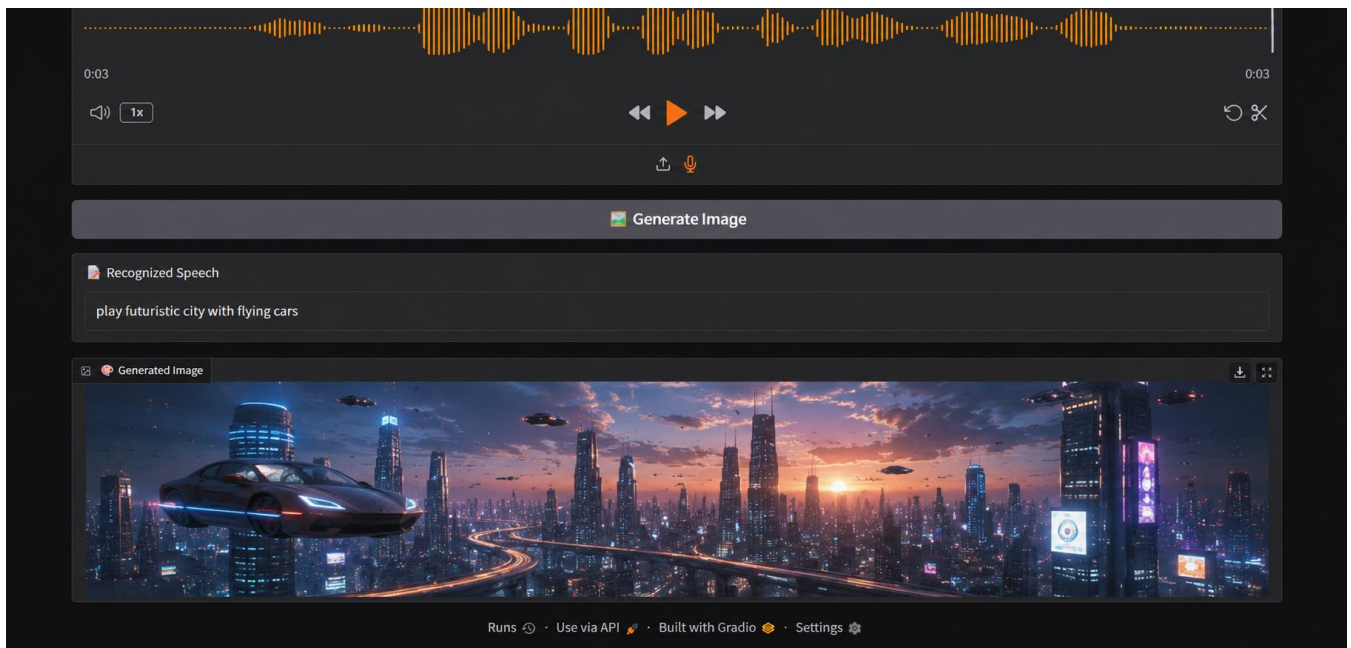
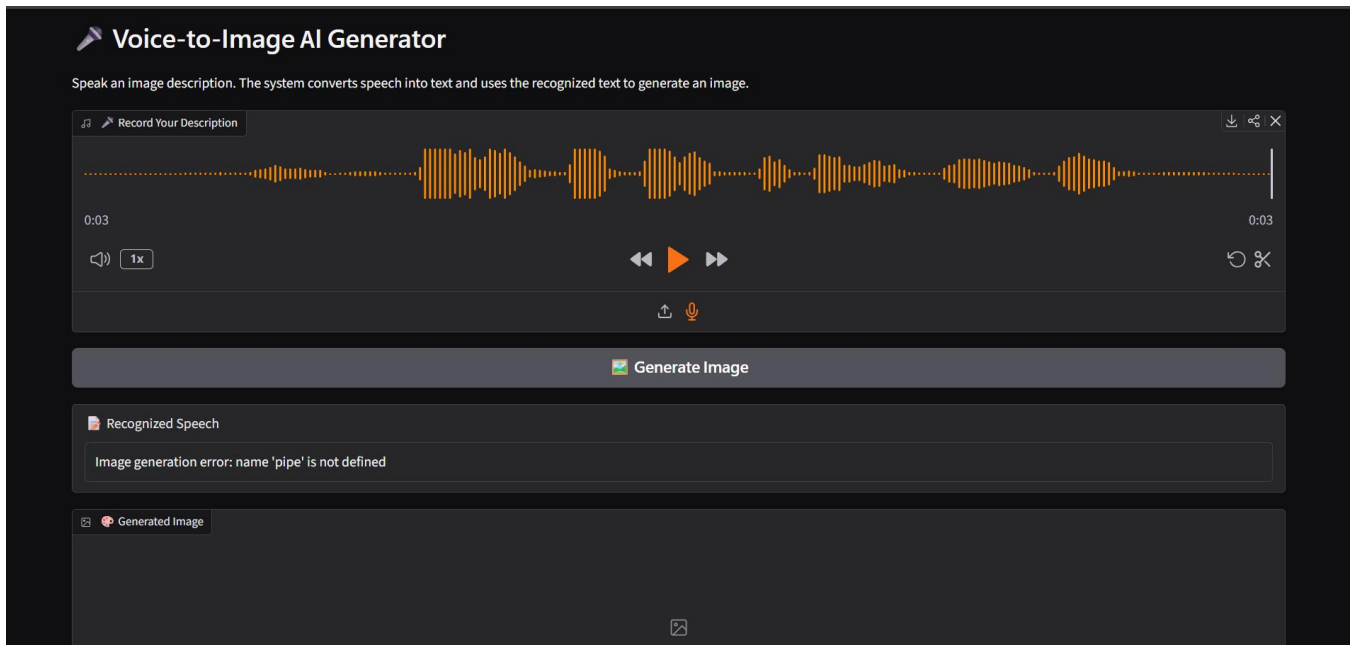
5. Workflow



6. Implementation

The application uses the SpeechRecognition library to convert the user's voice into text. The recognized text is checked for errors and then passed to the Stable Diffusion model as an image prompt.

A Gradio interface provides microphone/audio input and displays both the recognized speech and the generated image.



7. Sample Inputs

The user can speak descriptions such as:

- “A futuristic city with flying cars at night.”
- “A beautiful waterfall surrounded by green mountains.”
- “A robot studying in a modern classroom.”
- “A futuristic airport with advanced security robots.”

Example:

Voice:

"A futuristic city with flying cars"

↓

Recognized Speech:

A futuristic city with flying cars

↓

Generated Image:

[AI generated image]

8. Error Handling

The application handles basic unexpected inputs such as:

- No audio provided
- Speech that cannot be understood
- Speech recognition service errors
- Image generation errors

Appropriate error messages are displayed instead of terminating the application unexpectedly.

9. Real-World Application

This system can be useful for **voice-controlled creative design**, where users can describe an idea verbally and receive a visual representation without manually typing an image prompt.

10. Strengths, Limitations and Conclusion

Strengths

- Provides hands-free image generation.
- Combines speech recognition and image generation.
- Simple interface for users.
- Converts natural speech directly into visual content.

Limitations

- Speech recognition may fail with unclear audio or background noise.
- Image quality depends on the image-generation model.
- Generation can require significant GPU resources.
- Incorrectly recognized speech can produce an unrelated image.

Conclusion

The Voice-to-Image application successfully demonstrates an integrated multimodal Generative AI pipeline. Spoken input is converted into text and automatically used as a prompt for image generation. This demonstrates the integration of **speech and image-generation models** in a single application. The assessment expects the complete multimodal workflow and clear explanation of the tools and their limitations.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Pipeline Implementation	30%	Correctly implements a working text-to-image AND speech-to-text/text-to-speech pipeline	Both implemented with minor issues	Only one modality implemented correctly	Neither modality functions correctly
Output Quality & Relevance	25%	Generated outputs are high-quality and clearly relevant to inputs	Outputs are mostly relevant	Outputs are of inconsistent quality	Outputs are largely irrelevant or poor quality
Input Robustness	20%	Handles varied inputs robustly (different prompts/audio clips)	Handles common inputs well	Handles only the simplest inputs	Fails on basic inputs
Demo Polish	15%	Polished, engaging demo of both modalities	Clear demo with minor polish issues	Basic or rough demo	No usable demo presented
Tool/Model Explanation	10%	Clear explanation of models/tools used and their limitations	Adequate explanation	Minimal explanation	No explanation provided

Marks Evaluation:

Question No.	Pipeline Implementation (30%)	Output Quality & Relevance (25%)	Input Robustness (20%)	Demo Polish (15%)	Tool/Model Explanation (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						